

## ECN 164 International Finance — Practice Mock Exam A

Answer Key · Unofficial self-study practice based on Assignment 1, Part I (Chapter 2: quotes, cross rates, bid–ask spreads, triangular arbitrage)

**1. (5 points) Flipping a quote.** On October 2, 2026 the Australian dollar was worth about USD0.6550/AUD and the Canadian dollar about USD0.7250/CAD.

- How many Australian dollars does one U.S. dollar buy (AUD/USD)?
- How many Canadian dollars does one U.S. dollar buy (CAD/USD)?
- Which currency is the “base” currency in each of the original quotes?

**Solution**

- $\text{AUD/USD} = 1 / 0.6550 = \text{AUD1.5267/USD}.$
- $\text{CAD/USD} = 1 / 0.7250 = \text{CAD1.3793/USD}.$
- The base currency is the one being priced (in the denominator): the **AUD** in USD/AUD and the **CAD** in USD/CAD. The USD is the quote (pricing) currency in both.

**2. (5 points) Paying in a foreign currency.** You are traveling in South Korea and your hotel costs ₩180,000 (KRW) per night. The exchange rate is KRW1,380/USD.

- What is the cost of one night in dollars?
- A week later the rate is KRW1,395/USD. Did the won appreciate or depreciate against the dollar? What does the same room cost in dollars now?

**Solution**

- $180,000 / 1,380 = \text{\$130.43}$  per night.
- One dollar now buys more won ( $1,380 \rightarrow 1,395$ ), so the won **depreciated** against the dollar. New cost:  $180,000 / 1,395 = \text{\$129.03}$  — the room is cheaper for a dollar holder.

**3. (5 points) Nordic Furniture Imports, Inc.** needs to buy 5,000,000 Swedish kronor (SEK) to pay its Swedish supplier. Its banker quotes bid–ask rates of SEK9.4150–9.4200/USD. What will be the dollar cost of the SEK5,000,000?

**Solution**

The quote prices the dollar: the bank buys USD at SEK9.4150 (bid) and sells USD at SEK9.4200 (ask). The company buys SEK by *selling* dollars to the bank, so it gets the bank's bid for dollars — the rate that gives the company fewer kronor per dollar.

Dollar cost =  $5,000,000 / 9.4150 = \text{\$531,067.45}.$

(Check: using 9.4200 would give \$530,785.56, a lower cost — the bank never gives the customer the better rate.)

4. (7 points) **Rates of change.** The Australian dollar was USD0.6620/AUD on September 25, 2026 and USD0.6550/AUD one week later.

- By what percentage did the Australian dollar appreciate or depreciate against the dollar?
- By what percentage did the U.S. dollar appreciate or depreciate against the Australian dollar? (Hint: first express the rates as AUD/USD.)
- Why are your answers to (a) and (b) not exactly the same size?

**Solution**

- $(0.6550 - 0.6620) / 0.6620 = -1.06\%$ : the AUD **depreciated** by 1.06%.
- AUD/USD:  $1/0.6620 = 1.5106$  and  $1/0.6550 = 1.5267$ . Change =  $(1.5267 - 1.5106) / 1.5106 = +1.07\%$ : the USD **appreciated** by 1.07%.
- Each percentage change is measured relative to a different base (the starting price of a different currency). Since USD/AUD and AUD/USD are reciprocals,  $1 + \%\Delta(\text{AUD/USD}) = 1 / (1 + \%\Delta(\text{USD/AUD}))$ , which is not linear, so the two changes are close but not equal in size.

5. (8 points) **Cross rates.** Use the following quotes against the U.S. dollar: USD1.16/EUR, USD0.655/AUD, CNY7.12/USD, SEK9.40/USD.

- Calculate the cross rate between the Australian dollar and the euro, in AUD/EUR and in EUR/AUD.
- Calculate the yuan–euro cross rate (CNY/EUR).
- Calculate the Swedish krona–Australian dollar cross rate (SEK/AUD).
- Explain in one or two sentences why the dollar is called a “vehicle currency.”

**Solution**

- $\text{AUD/EUR} = (\text{USD/EUR}) / (\text{USD/AUD}) = 1.16 / 0.655 = \mathbf{1.7710}$ ;  $\text{EUR/AUD} = 0.655 / 1.16 = \mathbf{0.5647}$ .
- $\text{CNY/EUR} = (\text{CNY/USD}) \times (\text{USD/EUR}) = 7.12 \times 1.16 = \mathbf{8.2592}$ .
- $\text{SEK/AUD} = (\text{SEK/USD}) \times (\text{USD/AUD}) = 9.40 \times 0.655 = \mathbf{6.1570}$ .
- Most currencies trade actively only against the dollar, so a trade between two other currencies (e.g., AUD into SEK) is usually done in two legs through the dollar. The dollar is the “vehicle” that carries the trade, because its markets are the most liquid and cheapest to use.

**6. (7 points) Bid–ask spreads in the interbank market.** A dealer quotes the British pound at USD1.3318–1.3322/GBP.

- Which rate is the bid and which is the ask? Which rate do you get if you buy pounds from the dealer? If you sell pounds to the dealer?
- What is the spread in pips? What is the spread as a percentage of the ask price?
- A company buys GBP2,000,000 from the dealer and, after changing its mind, sells the pounds back to the dealer a minute later at the same quotes. How many dollars does the company lose?

**Solution**

- Bid = 1.3318 (the dealer buys GBP), ask = 1.3322 (the dealer sells GBP). You **buy** pounds at the ask, USD1.3322; you **sell** pounds at the bid, USD1.3318.
- Spread =  $1.3322 - 1.3318 = 0.0004 = 4 \text{ pips}$ . Percentage spread =  $0.0004 / 1.3322 = 0.0300\%$ .
- Pays  $2,000,000 \times 1.3322 = \$2,664,400.00$ ; receives  $2,000,000 \times 1.3318 = \$2,663,600.00$ . Loss = **\$800.00** ( $= \text{GBP}2,000,000 \times 0.0004$ ).

**7. (6 points) Bid–ask spreads at the airport.** At the airport, a currency kiosk buys pounds at USD1.25/GBP and sells pounds at USD1.40/GBP.

- What is the percentage spread? Compare it with your answer to Problem 6(b).
- You exchange \$1,500 into pounds at the kiosk, then change your mind and exchange all the pounds back into dollars. How many dollars do you end up with?
- Give two reasons why spreads for cash (“physical exchange”) are so much larger than interbank spreads.

**Solution**

- $(1.40 - 1.25) / 1.40 = 10.71\%$ , roughly 357 times the interbank spread of 0.0300%.
- Buy pounds at the ask:  $1,500 / 1.40 = \text{GBP}1,071.43$ . Sell them at the bid:  $1,071.43 \times 1.25 = \text{\$}1,339.29$  (a loss of \$160.71).
- Any two of: (1) cash transactions are small, so the fixed cost of each trade (staff, rent at the airport, the kiosk's own profit) is spread over few dollars; (2) physical currency must be stored, shipped, insured and protected, which is costly; (3) cash trading is much less liquid/competitive than the interbank market (travelers have few alternatives at the airport), so dealers face more inventory risk and can charge more.

**8. (5 points) Flipping a bid–ask quote.** A bank quotes the dollar at CHF0.7995–0.8005/USD. What are the bank's bid and ask prices for the Swiss franc, quoted in USD/CHF? (Hint: remember that the bank always buys low and sells high.)

**Solution**

When the bank buys CHF it is selling USD, which it does at its USD ask (CHF0.8005 per dollar); when it sells CHF it is buying USD at its USD bid (CHF0.7995).

Bid for CHF =  $1 / 0.8005 = \text{USD}1.24922/\text{CHF}$ ; ask for CHF =  $1 / 0.7995 = \text{USD}1.25078/\text{CHF}$ .

Check: bid < ask, so the bank still buys low and sells high.

**9. (6 points) Cross rates with bid–ask spreads.** A bank quotes USD1.3318–1.3322/GBP and CHF0.7995–0.8005/USD. A customer wants to trade pounds for Swiss francs, and the bank will go through the dollar.

a. What is the bank's bid rate for pounds in terms of francs (CHF/GBP)?

b. What is the bank's ask rate for pounds in terms of francs (CHF/GBP)?

***Solution***

a. Bank buys GBP: the customer sells GBP for USD at the GBP bid (1.3318), then sells those USD for CHF at the USD bid (0.7995). CHF/GBP bid =  $1.3318 \times 0.7995 = \mathbf{1.064774}$ .

b. Bank sells GBP: the customer buys USD with CHF at the USD ask (0.8005), then buys GBP at the GBP ask (1.3322). CHF/GBP ask =  $1.3322 \times 0.8005 = \mathbf{1.066426}$ .

**10. (6 points)** A Swiss watch company must pay GBP750,000 to a British supplier. It uses the bank and the quotes in Problem 9.

a. Which CHF/GBP rate applies, and how many Swiss francs will the payment cost?

b. How many more francs does the company pay than it would at the mid-rates of both quotes?

***Solution***

a. The company buys pounds from the bank, so it pays the bank's **ask** of CHF1.066426/GBP. Cost =  $750,000 \times 1.066426 = \mathbf{CHF799,819.57}$ .

b. Mid-rates: USD1.3320/GBP and CHF0.8000/USD give CHF1.065600/GBP, so the cost would be CHF799,200.00. Extra cost from the two spreads = **CHF619.57**.

**11. (7 points) Cross rates over time.** On September 1, 2026 the quotes were USD1.1500/EUR and CNY7.1500/USD. On October 1, 2026 they were USD1.1700/EUR and CNY7.1000/USD.

a. Calculate the CNY/EUR cross rate on both dates.

b. By what percentage did the euro appreciate or depreciate against the yuan?

c. By what percentage did the yuan appreciate or depreciate against the euro?

***Solution***

a. Sept 1:  $1.1500 \times 7.1500 = \mathbf{CNY8.2225/EUR}$ . Oct 1:  $1.1700 \times 7.1000 = \mathbf{CNY8.3070/EUR}$ .

b.  $(8.3070 - 8.2225) / 8.2225 = \mathbf{+1.03\%}$ : the euro **appreciated** against the yuan.

c. EUR/CNY:  $1/8.2225 = 0.121618$  and  $1/8.3070 = 0.120380$ . Change = **-1.02%**: the yuan **depreciated** by 1.02% against the euro.

**12.** (7 points) As a foreign exchange trader, you see the following quotes for Australian dollars (AUD), U.S. dollars (USD), and Japanese yen (JPY):

USD0.6550/AUD JPY99.20/AUD JPY148.60/USD

Is there an arbitrage opportunity, and if so, how would you exploit it? Compute your profit per \$1,000,000.

**Solution**

Implied JPY/AUD = (JPY/USD) × (USD/AUD) = 148.60 × 0.6550 = 97.3330. The quoted rate is JPY99.20/AUD > 97.3330, so **yes**: the AUD is too expensive in the yen market (equivalently, yen are cheap there).

Exploit it by buying AUD where it is cheap (with dollars) and selling it where it is expensive (for yen):

1) USD → AUD: 1,000,000 / 0.6550 = AUD1,526,717.56

2) AUD → JPY: × 99.20 = JPY151,450,381.68

3) JPY → USD: / 148.60 = \$1,019,181.57

Profit = **\$19,181.57** (1.92%).

**13.** (10 points) **Triangular arbitrage.** You observe the following quotes (ignore bid–ask spreads): USD0.6550/AUD in New York, USD0.7250/CAD in Toronto, and CAD0.9200/AUD in Sydney.

- What CAD/AUD cross rate is implied by the two dollar quotes?
- Is there an arbitrage opportunity? In which market is the Australian dollar “too expensive”?
- You start with \$1,000,000. Describe the three trades you would make and calculate your profit in dollars.
- What will happen to the three exchange rates as many traders do the same thing?

**Solution**

a. CAD/AUD = (USD/AUD) / (USD/CAD) = 0.6550 / 0.7250 = **0.9034**.

b. Yes. Sydney quotes CAD0.9200/AUD > 0.9034, so the AUD is **too expensive in Sydney** (it buys more CAD there than it should).

c. 1) New York: buy AUD with USD: 1,000,000 / 0.6550 = AUD1,526,717.56. 2) Sydney: sell AUD for CAD: 1,526,717.56 × 0.9200 = CAD1,404,580.15. 3) Toronto: sell CAD for USD: 1,404,580.15 × 0.7250 = \$1,018,320.61. Profit = **\$18,320.61**.

d. Buying AUD in New York pushes USD/AUD **up**; selling AUD for CAD in Sydney pushes CAD/AUD **down**; selling CAD for USD in Toronto pushes USD/CAD **down**. The implied cross rate (USD/AUD)/(USD/CAD) rises while the Sydney rate falls, until they are equal and the profit disappears.

**14. (10 points) Triangular arbitrage with bid–ask spreads.** Now the quotes are: USD0.6548–0.6552/AUD, USD0.7248–0.7252/CAD, and CAD0.9020–0.9030/AUD. Starting with \$1,000,000, check both possible routes (dollars → Australian dollars → Canadian dollars → dollars, and dollars → Canadian dollars → Australian dollars → dollars). Is there an arbitrage profit once you pay the bid–ask spreads?

***Solution***

**Route 1: USD → AUD → CAD → USD**

Buy AUD at the ask:  $1,000,000 / 0.6552 = \text{AUD}1,526,251.53$ . Sell AUD for CAD at the CAD/AUD bid:  $\times 0.9020 = \text{CAD}1,376,678.88$ . Sell CAD at the USD/CAD bid:  $\times 0.7248 = \text{\$}997,816.85$  (loss of \$2,183.15).

**Route 2: USD → CAD → AUD → USD**

Buy CAD at the ask:  $1,000,000 / 0.7252 = \text{CAD}1,378,929.95$ . Buy AUD with CAD at the CAD/AUD ask:  $/ 0.9030 = \text{AUD}1,527,054.21$ . Sell AUD at the USD/AUD bid:  $\times 0.6548 = \text{\$}999,915.10$  (loss of \$84.90).

**No arbitrage profit.** Both routes lose money. Equivalently, the implied CAD/AUD bid–ask is  $0.9029\text{--}0.9040$  ( $= 0.6548/0.7252$  to  $0.6552/0.7248$ ), which overlaps the quoted  $0.9020\text{--}0.9030$ , so the spreads eat any gain.

**15. (6 points) The foreign exchange market.** Answer briefly.

- What is “liquidity,” and why do the most-traded currency pairs (such as EUR/USD) have smaller bid–ask spreads than rarely traded pairs (such as SEK/MXN)?
- What is the interbank market, and how does it differ from the retail market in which tourists and small firms buy currency?
- A dealer does not try to predict exchange rates. How does the dealer still make a profit?

***Solution***

- Liquidity is the ability to buy or sell a large amount quickly without moving the price. In heavily traded pairs, dealers can offset positions almost immediately and fixed costs are spread over huge volume, and competition among many dealers squeezes spreads; thin pairs carry more inventory risk, so dealers charge wider spreads.
- The interbank (wholesale) market is where large banks and dealers trade with each other in very large amounts (often millions of dollars) electronically or by phone at very narrow spreads. The retail market involves small transactions with end customers at much wider spreads.
- By buying at the bid and selling at the ask: the spread between the two prices is the dealer's compensation for providing immediacy and bearing inventory risk.